

Scientific Computing & Advanced Programming Lab

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Lecture-Slides-7



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Eigenvalue Decomposition

- ▶ For a square matrix $A \in \mathbb{R}^{n \times n}$, an eigenvector $\mathbf{v} \neq \mathbf{0}$ satisfies

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ is the corresponding eigenvalue.

- ▶ If A has n linearly independent eigenvectors, we can write

$$A = V\Lambda V^{-1} \text{ where}$$

$$V = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix}, \quad \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

Interpretation

The eigenvectors provide special directions that are only **scaled** by the action of A .



Why Eigen decomposition?

- ▶ If $A = V\Lambda V^{-1}$, then many matrix operations become simpler.
- ▶ For example, $A^k = V\Lambda^k V^{-1}$, since Λ is diagonal,

$$\Lambda^k = \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{bmatrix}.$$

- ▶ This is useful for studying: powers of matrices, dynamical systems, stability, differential equations, iterative numerical methods, etc. .
- ▶ If A is symmetric i.e., $A = A^T$, the decomposition has the particularly nice form

$$A = Q\Lambda Q^T$$

where Q is orthogonal: $Q^T Q = I$.



Computing eigenvalues exactly is "impossible."

- ▶ There is no direct method for computing eigenvalues for matrices of size five or higher in general
- ▶ One may wonder if some day we will discover a direct method.
- ▶ It can be proved that such a method cannot exist for general matrices of size five or higher.
- ▶ The Abel-Ruffini theorem (also known as Abel's impossibility theorem) says that there is no general algebraic solution to polynomial equations of degree five or higher.
- ▶ no direct method exists for exactly finding roots of polynomials of degree five or higher.



Methods for computing a single eigenvalue: Power Iteration

- ▶ Suppose that A is a square diagonalizable matrix.
- ▶ Then A has an eigenvalue decomposition $A = V\Lambda V^{-1}$
- ▶ Assume that the eigenvectors form a basis. Write an arbitrary initial vector x_0 as:

$$x_0 = c_1 v_1 + c_2 v_2 + \dots + c_n v_n.$$

- ▶ Applying A :

$$Ax_0 = c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \dots + c_n \lambda_n v_n.$$

- ▶ Similarly, applying A k times we have:

$$A^k x_0 = c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2 + \dots + c_n \lambda_n^k v_n.$$

- ▶ Suppose: $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|.$



Methods for computing a single eigenvalue: Power Iteration

► Then:

$$A^k x_0 = \lambda_1^k \left[c_1 v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k v_2 + \dots + c_n \left(\frac{\lambda_n}{\lambda_1} \right)^k v_n \right]$$

► As $k \rightarrow \infty$:

$$\frac{|\lambda_i|^k}{|\lambda_1|^k} \rightarrow 0, \quad i > 1$$

► Therefore, the first term in the sum dominates the sum as $k \rightarrow \infty$

► Consequently,

$$A^k x_0 \approx \lambda_1^k c_1 v_1$$



Power Iteration: Algorithm

Given: $A \in \mathbb{R}^{n \times n}$, tolerance $\varepsilon > 0$

1. Choose a nonzero initial vector x_0 and normalize:

$$x_0 \leftarrow \frac{x_0}{\|x_0\|_2}.$$

2. For $k = 0, 1, 2, \dots$:

$$y_k = Ax_k,$$

$$x_{k+1} = \frac{y_k}{\|y_k\|_2}.$$

3. Check convergence:

$$\|x_{k+1} - x_k\|_2 < \varepsilon.$$



Power Iteration: Algorithm

- ▶ The last algorithm only gives the dominant eigenvector.
- ▶ What about the eigenvalue?

Rayleigh quotient iteration

- ▶ if x is an eigenvector with $Ax = \lambda x$
- ▶ Then $x^T Ax = \lambda x^T x$
- ▶ Define $r(x)$ as:
$$r(x) = \frac{x^T Ax}{x^T x}, \quad x \neq 0$$
- ▶ Every iteration on the power iteration compute:

$$r_k = \frac{x_k^T A x_k}{x_k^T x_k}$$



Rayleigh Quotient

- ▶ For $x \neq 0$, the Rayleigh quotient is $r(x) = \frac{x^T A x}{x^T x}$.
- ▶ For a symmetric matrix $A = A^T$,

$$\nabla r(x) = \frac{2}{x^T x} (Ax - r(x)x).$$

- ▶ If x_i is an eigenvector, $Ax_i = \lambda_i x_i$, then

$$r(x_i) = \lambda_i$$

- ▶ and therefore $\nabla r(x_i) = 0$.

- ▶ Thus, eigenvectors are stationary points of the Rayleigh quotient.



Local Error of the Rayleigh Quotient

- ▶ Let x_i be an eigenvector and let y be close to x_i .
- ▶ Taylor expansion:

$$r(y) = r(x_i) + \nabla r(x_i)^T (y - x_i) + \frac{1}{2} (y - x_i)^T H(x_i) (y - x_i) + \mathcal{O}(\|y - x_i\|^3).$$

- ▶ Since $\nabla r(x_i) = 0$, $r(x_i) = \lambda_i$, we obtain

$$r(y) - \lambda_i = \frac{1}{2} (y - x_i)^T H(x_i) (y - x_i) + \mathcal{O}(\|y - x_i\|^3).$$

- ▶ Hence, for $A = A^T$ $\boxed{|r(y) - \lambda_i| = \mathcal{O}(\|y - x_i\|^2)}$
- ▶ In other words, the Rayleigh quotient has second-order eigenvalue accuracy.



Numerical Verification of Second-Order Accuracy

▶
$$d_k \approx C e_k^2, \quad e_k = \|x_k - x_i\|, \quad d_k = |r(x_k) - \lambda_i|,$$

▶
$$\log d_k \approx \log C + 2 \log e_k.$$

▶ For two successive iterations,

$$\log d_{k+1} - \log d_k \approx 2(\log e_{k+1} - \log e_k).$$

▶ Hence define the observed order

$$p_k = \frac{\log d_{k+1} - \log d_k}{\log e_{k+1} - \log e_k} = \frac{\log(d_{k+1}/d_k)}{\log(e_{k+1}/e_k)} \longrightarrow 2.$$



What Changes for a Nonsymmetric Matrix?

► For general A , $r(x) = \frac{x^T A x}{x^T x}$ still makes sense.

► However, $\nabla(x^T A x) = (A + A^T)x$, so

$$\nabla r(x) = \frac{(A + A^T)x - 2r(x)x}{x^T x}.$$

► If $Ax_i = \lambda_i x_i$, then $\nabla r(x_i) = \frac{(A^T - A)x_i}{x_i^T x_i}$, which is generally non-zero.

► Therefore, the stationary-point argument is special to $A = A^T$.

► The Rayleigh quotient can still estimate eigenvalues, but its second-order error result does not generally hold for nonsymmetric matrices.



QR Decomposition: Motivation

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix}$$

a matrix whose columns are not necessarily orthogonal.

- ▶ We would like to replace the columns of A by an **orthonormal basis** spanning the same space.

$$\boxed{\{a_1, \dots, a_n\} \longrightarrow \{q_1, \dots, q_n\}}$$

where

$$q_i^T q_j = \delta_{ij}.$$

- ▶ This leads to the factorization

$$\boxed{A = QR}$$

where Q contains the orthonormal basis and R contains the



Orthogonal and Orthonormal Vectors

- ▶ Two vectors q_i and q_j are orthogonal if

$$q_i^T q_j = 0, \quad i \neq j.$$

- ▶ They are orthonormal if, in addition,

$$\|q_i\| = 1.$$



$$Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{pmatrix},$$

then orthonormality of the columns is equivalent to

$$\boxed{Q^T Q = I} \implies \boxed{Q^{-1} = Q^T}$$

- ▶ An important consequence is $\|Qx\| = \|x\|$. Thus orthogonal matrices preserve lengths and angles.



Gram–Schmidt: First Step

Given linearly independent vectors $a_1, a_2, \dots, a_n$, we construct an orthonormal set $q_1, q_2, \dots, q_n$.

1. Start with the first vector:

$$\boxed{q_1 = \frac{a_1}{\|a_1\|}} \implies \|q_1\| = 1.$$

2. The second vector a_2 generally has a component in the q_1 direction. Its projection onto q_1 is

$$\text{proj}_{q_1}(a_2) = (q_1^T a_2)q_1.$$

3. Remove this component: $u_2 = a_2 - (q_1^T a_2)q_1$.

4. Then

$$q_1^T u_2 = 0.$$



Gram–Schmidt: Normalization

- ▶ We now normalize the orthogonalized vector:

$$q_2 = \frac{u_2}{\|u_2\|}$$

- ▶ Therefore, $q_1^T q_2 = 0$, $\|q_1\| = \|q_2\| = 1$.

- ▶ For the third vector, remove its components in both previous directions:

$$u_3 = a_3 - (q_1^T a_3)q_1 - (q_2^T a_3)q_2.$$

- ▶ Then $q_3 = \frac{u_3}{\|u_3\|}$.

The general step is therefore

$$u_j = a_j - \sum_{i=1}^{j-1} (q_i^T a_j) q_i$$

followed by

$$q_j = \frac{u_j}{\|u_j\|}.$$



From Gram-Schmidt $\rightarrow$ QR

- ▶ Rearranging the Gram-Schmidt equations gives

$$a_1 = (q_1^T a_1)q_1,$$

$$a_2 = (q_1^T a_2)q_1 + (q_2^T a_2)q_2,$$

- ▶ and, in general, $a_j = \sum_{i=1}^j (q_i^T a_j)q_i.$

- ▶ Thus every column of A can be expressed in the orthonormal basis $q_1, \dots, q_n.$

- ▶ Collecting the q_i into $Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{pmatrix},$ we obtain

$$A = QR.$$



Structure of R

- ▶ The coefficients in the expansion of a_j are $R_{ij} = q_i^T a_j$, $i \leq j$ and for $i > j$, $R_{ij} = 0$.

- ▶ Therefore $R = \begin{pmatrix} * & * & * & \cdots \\ 0 & * & * & \cdots \\ 0 & 0 & * & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$ is upper triangular.

- ▶ Hence $A = QR$, $Q^T Q = I$, R upper triangular.



QR Decomposition: Example

Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

Its columns are $a_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $a_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

First,

$$q_1 = \frac{a_1}{\|a_1\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The projection of a_2 onto q_1 is

$$(q_1^T a_2)q_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore

$$u_2 = a_2 - (q_1^T a_2)q_1 = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$



QR Decomposition: Example Continued

Normalize u_2 : $q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Therefore

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Now compute

$$R_{ij} = q_i^T a_j.$$

Hence

$$R = \begin{pmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{pmatrix}.$$



QR Decomposition: Pseudocode

Input: $A = [a_1, \dots, a_n] \in \mathbb{R}^{m \times n}$

Output: $Q = [q_1, \dots, q_n]$ and R

1. For $j = 1, \dots, n$:
2. Start with the current column: $u_j \leftarrow a_j$
3. Remove its components in the previous directions:
For $i = 1, \dots, j - 1$:

$$R_{ij} \leftarrow q_i^T a_j, \quad u_j \leftarrow u_j - R_{ij} q_i$$

4. Normalize the remaining component:

$$R_{jj} \leftarrow \|u_j\|, \quad q_j \leftarrow \frac{u_j}{R_{jj}}$$



From QR Decomposition to Eigenvalues

- ▶ We now have a way to factor any matrix A as $A = QR$, where Q is orthogonal and R is upper triangular.
- ▶ Can repeated QR factorizations help us find eigenvalues?
- ▶ Recall the power iteration:

$$Ax_k \longrightarrow \lambda_1 x_k.$$

It repeatedly applies A and extracts the dominant eigendirection.

- ▶ To find *all* eigenvalues, we need to apply the same idea to several directions simultaneously.



From One Vector to a Matrix

- ▶ Power iteration starts with one vector: x_0 .
- ▶ Instead, start with an orthonormal basis:

$$X_0 = \begin{pmatrix} | & & | \\ x_1 & \cdots & x_n \\ | & & | \end{pmatrix}, \quad X_0^T X_0 = I.$$

- ▶ Apply A to every column: $Y_1 = AX_0$. After repeated multiplication,

$$Y_k = A^k X_0.$$

- ▶ This is essentially power iteration applied to **all directions simultaneously**.

- ▶ But there is a problem: $X_0^T X_0 = I \not\Rightarrow (AX_0)^T (AX_0) = I$.

- ▶ The columns lose orthogonality.



Orthogonal Iteration

- ▶ After applying A , restore orthonormality using QR decomposition.
- ▶ Starting with X_k , $AX_k = Q_k R_k$. Set $X_{k+1} = Q_k$.
- ▶ Thus one iteration is $X_k \xrightarrow{A} AX_k \xrightarrow{\text{QR}} X_{k+1}$.
- ▶ This gives the **orthogonal iteration**.
- ▶ It is the matrix analogue of power iteration:

Power iteration = multiply + normalize

Orthogonal iteration = multiply + orthogonalize.



Why Does Orthogonal Iteration Work?

- ▶ Suppose A has orthonormal eigenvectors: $AV = V\Lambda$, where $V^T V = I$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- ▶ Write the initial basis as $X_0 = VC$. Then $A^k X_0 = V\Lambda^k C$.
- ▶ As in power iteration, the columns become increasingly aligned with the dominant eigendirections.
- ▶ But instead of allowing the columns to become parallel, we repeatedly orthogonalize them.
- ▶ Thus the different columns separate into different eigendirections.

$$X_k \longrightarrow V$$

up to signs and ordering, under suitable assumptions.



Starting Point: Orthogonal Iteration

- ▶ start with $X_0 = I$.
- ▶ The orthogonal iteration applies A to the current basis and orthogonalizes: $AX_k = X_{k+1}\tilde{R}_k$.
- ▶ Thus $AX_0 \xrightarrow{\text{QR}} X_1\tilde{R}_0$, $AX_1 \xrightarrow{\text{QR}} X_2\tilde{R}_1$, $\dots$
- ▶ Since $X_0 = I$, $AX_0 = A$.
- ▶ Compute the first QR decomposition: $A = Q_0R_0$.
- ▶ Therefore $X_1 = Q_0$.

Question

Can we avoid repeatedly multiplying the original A by the increasingly complicated matrices X_k ?



Iteration 1: From X_1 to X_2

- ▶ We currently have $\boxed{X_1 = Q_0}$: $AX_1 = AQ_0$.
- ▶ Since $A = Q_0 R_0$, we obtain $AX_1 = Q_0 R_0 Q_0$.
- ▶ Rearrange: $AX_1 = Q_0 \underbrace{(R_0 Q_0)}_{\text{new matrix}}$.
- ▶ This suggests defining $\boxed{A_1 = R_0 Q_0}$.
- ▶ Hence $\boxed{AX_1 = X_1 A_1}$.
- ▶ Now factor the *new matrix* : $\boxed{A_1 = Q_1 R_1}$.
- ▶ Therefore: $AX_1 = X_1 Q_1 R_1$.
- ▶ Thus the next orthogonal factor is:

$$\boxed{X_2 = X_1 Q_1 = Q_0 Q_1}.$$



Iteration 2: From X_2 to X_3

► We now have: $X_2 = Q_0 Q_1$, $A_1 = Q_1 R_1$.

► Orthogonal iteration requires AX_2 .

► Since $X_2 = X_1 Q_1$, we have

$$AX_2 = AX_1 Q_1.$$

► From the previous iteration, $AX_1 = X_2 R_1$. Therefore

$$AX_2 = X_2 R_1 Q_1.$$

► Define $A_2 = R_1 Q_1$. Hence $AX_2 = X_2 A_2$.

► Now factor A_2 : $A_2 = Q_2 R_2$.

► Thus $AX_2 = X_2 Q_2 R_2$.

► Therefore $X_3 = X_2 Q_2 = Q_0 Q_1 Q_2$.



Iteration 3: From X_3 to X_4

We now have

$$\boxed{X_3 = Q_0 Q_1 Q_2}, \quad \boxed{A_2 = Q_2 R_2}.$$

Again, orthogonal iteration requires

$$AX_3.$$

Since

$$X_3 = X_2 Q_2,$$

we obtain

$$AX_3 = AX_2 Q_2.$$

From the previous iteration,

$$AX_2 = X_3 R_2.$$

Hence

$$AX_3 = X_3 R_2 Q_2.$$

Define



Choosing $X_0 = I$

- ▶ Now make the particular choice $X_0 = I$.
- ▶ The first orthogonal iteration is $AX_0 = A$.
- ▶ Compute its QR decomposition: $A = Q_0R_0$.
- ▶ Therefore $X_1 = Q_0$.
- ▶ The next iteration gives $AX_1 = AQ_0$.
- ▶ Since $A = Q_0R_0$, we have $AQ_0 = Q_0R_0Q_0$.
- ▶ Therefore the important matrix

$$A_1 = R_0Q_0$$

appears naturally.



The QR Step

► At iteration k , factor $A_k = Q_k R_k$.

► Instead of continuing with AQ_k , define $A_{k+1} = R_k Q_k$.

► Because $R_k = Q_k^T A_k$, and therefore

$$A_{k+1} = R_k Q_k = Q_k^T A_k Q_k.$$

► Since Q_k is orthogonal, $Q_k^{-1} = Q_k^T \implies A_{k+1} = Q_k^{-1} A_k Q_k$.

► Therefore A_{k+1} and A_k are similar.

► Similar matrices have the same characteristic polynomial:
 $\det(\lambda I - A_{k+1}) = \det(\lambda I - A_k)$.

► Hence $\text{eig}(A_{k+1}) = \text{eig}(A_k)$.

► So QR iteration changes the matrix while **preserving all its eigenvalues**.



The Basic QR Algorithm

► Start with: $A_0 = A$.

► For $k = 0, 1, 2, \dots$:

$$A_k = Q_k R_k$$

$$A_{k+1} = R_k Q_k.$$

► Repeat until the matrix is approximately diagonal. For a symmetric matrix, monitor

$$\text{off}(A_k) = \left(\sum_{i \neq j} |(A_k)_{ij}|^2 \right)^{1/2}.$$

► Stop when $\text{off}(A_k) < \text{tol.}$

► Then $\lambda_i \approx (A_k)_{ii}.$



QR Iteration and Eigenvectors

- ▶ QR iteration also contains information about the eigenvectors.
- ▶ At each step, $A_k = Q_k R_k$. Define the accumulated orthogonal transformation

$$V_k = Q_0 Q_1 \cdots Q_{k-1}.$$

- ▶ Then

$$A_k = V_k^T A V_k.$$

- ▶ If QR iteration converges to

$$A_k \longrightarrow \Lambda,$$

then

$$V_k^T A V_k \longrightarrow \Lambda.$$

- ▶ Therefore the columns of V_k approach the eigenvectors:

$$V_k \longrightarrow V.$$



Singular Value Decomposition (SVD)

For any matrix $A \in \mathbb{R}^{m \times n}$, there exist orthogonal matrices

$$U \in \mathbb{R}^{m \times m}, \quad V \in \mathbb{R}^{n \times n},$$

and a diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$, such that

$$\boxed{A = U\Sigma V^T}.$$

The diagonal entries of Σ are the **singular values**

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

- ▶ Columns of V : right singular vectors
- ▶ Columns of U : left singular vectors
- ▶ σ_i : singular values



SVD and Eigenvalue Problems

Consider: $A^T A \in \mathbb{R}^{n \times n}$.

Since $A^T A$ is symmetric positive semidefinite,

$$A^T A v_i = \lambda_i v_i, \quad \lambda_i \geq 0.$$

The singular values are $\sigma_i = \sqrt{\lambda_i}$.

If $\sigma_i > 0$, define

$$u_i = \frac{A v_i}{\sigma_i}.$$

Then

$$A v_i = \sigma_i u_i, \quad A^T u_i = \sigma_i v_i.$$

Therefore,

$$A = U \Sigma V^T$$

SVD of $A \iff$ eigendecomposition of $A^T A$.

